

Motion Controller Programming Guide

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Communication Properties

Full Command Syntax : <TASK> <OBJECTIVE> <AXIS> <VALUE>

Total Characters expected < 64

Command termination: Carriage return followed by line feed '\r\n'

Baud rate: 9600

Timeout = 1 second (real-time configurable)

Device States

INIT : Microcontroller initialization state. Device will await until a valid device list to control is specified over the serial port.

IDLE : Microcontroller awaiting a command. Device will continuously check serial port for incoming data bits.

ACTIVE : Microcontroller is processing a move command. Currently, this state can never be fetched because move commands require complete control of program execution.

ERROR : Microcontroller is in an error state and cannot proceed to operate.

INIT Step	Microcontroller Message
Board Alias	Waiting for board name...
Board Validation	Validating board alias. Please wait.
Device list input	Waiting for motor list...
Motor Validation	Validating motor list. Please wait.
Completion	Device controller initialization complete.

Initialization

The board can be programmed at compile or runtime for a pre-defined microcontroller pinout.

Case 1 : Board defined at Compile Time :

The board can be programmed at compile or runtime for operation with a pre-defined motor class. Program and dynamic memory consumption is reduced to support ONLY the motors operated by the specified microcontroller.

Case 2 : Board defined at Runtime :

The board is programmed to support all pre-defined motor classes. During run-time, the microcontroller alias is used to configure the microcontroller properly.

Board Alias	Rotating Randy			
Dev Axis	Dev CS	Motor	Motor Axis	Motor CS
1	0	NEMA17	1	0
2	0	STEPPER_ONLINE	1	0
Board Alias	Precision Plunge			
Dev Axis	Dev CS	Motor	Motor Axis	Motor CS
1	0	G2	1	0
1	1	G2	1	1
Board Alias	Whirling William			
Dev Axis	Dev CS	Motor Alias	Motor Axis	Motor CS
1	0	STEPPER_ONLINE	1	0
1	1	STEPPER_ONLINE	1	1

Real-Time Commands

1. Device Initialization/Removal Commands:

- i. Configure for specific board :
 - a) Boards : { “ROTATING RANDY”, “PRECISION PLUNGE”, “WHIRLING WILLIAM” }
- ii. Open/Close Channel [3]
 - a) TASK = { “OPEN”, “CLOSE” }
 - b) OBJECTIVE = { “ALL_CHANNELS” [2], “CHANNEL” [3] }
 - c) EXAMPLE:
 - 1. CMD : “OPEN ALL_CHANNELS”
 - 2. CMD : “CLOSE CHANNEL” + <uint8_t channel>

2. Getters and Setters :

- i. TASK = { “GET” ([3] arguments), “SET” ([4] arguments) }
- ii. OBJECTIVE = { “EN”, “POS”, “VEL”, “ACC”, “ORIGIN”
 - a) Velocity :
 - 1. Rotational Device : rad/s
 - 2. Positional Device : mm/s

3. Stepper Motors :

- i. [3] CMD = “MOVE STEPS <int32_t>”

4. Special Commands :

- i. [1] CMD = “HELP” : Printout high level command syntax and commands
- ii. [2] CMD = “RESET CLIENT” : Software reset of the device
- iii. [2] CMD = “REMOVE CLIENT” : Device resets to startup state
- iv. [2] CMD = “DISPLAY COMMANDS” : List all supported commands
- v. [3] CMD = “SET TIMEOUT <SECONDS>” : Defines connection timeout of dev

Examples

1. **Initialize Microcontroller** with board alias *Rotating Randy*
Prompt: Waiting for board name...
CMD = "Rotating Randy"
Response : The controller is programmed for operating with the configuration of *Rotating Randy* or returns to the board alias prompt, if specifications are invalid.
2. **Reset Microcontroller**
CMD = "RESET CLIENT"
Response : Microcontroller reboots.
3. **Reset Microcontroller** board alias
CMD = "REMOVE CLIENT"
Response : Microcontroller resets to startup state.
4. **Open connection** with axis 2
CMD = "OPEN CHANNEL 2"
Response : Channel 2 begins startup configuration for operation.
5. **Close connection** with axis 2
CMD = "CLOSE CHANNEL 2"
Assume : Channel 2 begins disconnection procedure.
6. **Set Enable** output of axis 1.
CMD = "SET EN 1" + <int outputsOn>
Response : Axis 1 output is powered on or off (outputsOn == 0).\
7. **Get enable** status of axis 1
CMD = "GET EN 1"
Response : Axis 1 output enable status is returned
8. **Get rotation** of the stepper motor belonging to axis 1.
CMD = "GET POS 1"
Response : The rotation of axis 1 is returned (units can vary).
9. **Set velocity** of linear actuator belonging to axis 3
CMD = "SET VEL 3 100"
Response : Sets the velocity of axis 3 to 100 units.
10. **Set acceleration** of linear actuator belonging to axis 3
CMD = "SET ACC 3 5"
Response : Sets the velocity of axis 3 to 5 units.
11. **Set angular frequency** of rotator belonging to axis 1
CMD = "SET VEL 1 3.14"
Response : Sets the frequency of the rotator to 0.5 Hz

Board Pinouts :

Board Alias	Whirling William		
Pin	Pin Mode	Signal Cable	Designation
2	Digital OUT	Red	Enable
3	Digital OUT	Yellow	Pulse
4	Digital OUT	Orange	Direction
5	Digital OUT	White	CS
6	Digital OUT	Black	CLK
GND	X	Green	Ground
Board Alias	Precision Plunge		
Pin	Pin Mode	Signal Cable	Designation
A0	INPUT		M1: Current-Sense
A1	OUTPUT		M1: PWM
A4	INPUT_PULLUP		M1: Fault
A5	INPUT_PULLUP		M2: Fault
A8	OUTPUT		M2: PWM
A9	INPUT		M2: Current-Sense
1	Digital OUT		M1: Sleep
2	Digital OUT		M1: Direction
3	Digital OUT		M2: Sleep
4	Digital OUT		M2: Direction
5	Digital IN		M1 Hall Effect: I
6	Digital IN		M1 Hall Effect: Q
7	Digital IN		M2 Hall Effect: I
8	Digital IN		M2 Hall Effect: Q